PRAVIHARA NERANJANA ABEGUNASEKARA

Software Engineer | .NET & Full-Stack Developer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 1 year of professional experience at Factum Software Solutions, building and maintaining an enterprise multi-tenant SaaS platform for clients in Saudi Arabia. Final-year BSc (Hons) Software Engineering undergraduate at NSBM Green University with hands-on experience in C#, ASP.NET and SQL Server, RESTful API development, relational database design and query optimisation, and object-oriented programming. Experienced in building AI-enabled, data-driven applications, working in Agile teams, and using Git, GitHub, Docker and CI/CD. Seeking a Junior Software Engineer (.NET) role to grow in enterprise .NET development.

TECHNICAL SKILLS

Languages: C#, SQL, JavaScript, TypeScript, PHP, Python, Java, Dart

.NET and Backend: ASP.NET, RESTful API design and integration, Laravel, FastAPI, Node.js, Supabase Edge Functions

Databases: SQL Server, MySQL, PostgreSQL, Redis, Firebase; schema design, migrations, SQL aggregation and query optimisation

Frontend: HTML, CSS, JavaScript, React, Next.js, Inertia.js, Tailwind CSS, Vite, Flutter

Core Concepts: Object-oriented programming, client-server architecture, multi-tenant SaaS, role-based access control, performance optimisation

Cloud and DevOps: Git and GitHub, GitHub Actions, CI/CD pipelines, Docker, AWS (Translate)

AI and Data: Generative AI APIs, Google Gemini API, LLM integration, XGBoost, scikit-learn, SHAP, pandas, NumPy

Testing and Practices: pytest, Vitest, debugging and troubleshooting, Agile, internationalisation (English and Arabic, RTL)

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer – Factum Software Solutions (Pvt) Ltd, Sri Lanka
Nov 2025 – Present (joined as Software Engineering Intern, retained after internship)

Project: Operate – Multi-Tenant Operations and Compliance Management SaaS Platform (client base: Saudi Arabia)
Tech: Laravel 12, PHP, React, Inertia.js, Tailwind CSS, Vite, MySQL, Redis, AWS Translate, Generative AI APIs

- Optimised dashboard performance by rewriting heavy queries as SQL aggregations, reducing query load by 84 to 95 percent, and troubleshot and fixed three out-of-memory crashes before release.
- Upgraded the backend from Laravel 9 to Laravel 12, including dependency updates and compatibility fixes, keeping the production application maintainable.
- Implemented the AI-powered Knowledge Center, integrating Generative AI APIs to generate knowledge-base content, photos and videos from user prompts.
- Rebuilt the Announcements module end to end (backend and frontend): composer, scheduling, attachments, and read and acknowledgement tracking.
- Helped migrate the frontend from Bootstrap/Blade to React and Tailwind CSS using Inertia.js, moving about 130 pages to the new layout.
- Rebuilt the main dashboard in React with 13 customisable widgets (drag and drop, resize, show or hide, saved layouts).
- Debugged and resolved Arabic localisation and RTL layout issues in a bilingual (English and Arabic) application.
- Added dark mode, shared UI components and reusable React hooks to improve usability and code reuse.

PROJECTS

Auctinier – Online Vehicle and Spare Parts Auction System
C#, ASP.NET, SQL Server

- Second-year group project: an online auction management system for vehicles and vehicle spare parts.
- Developed the application with ASP.NET and C# using object-oriented design, backed by a SQL Server relational database to manage vehicle and spare-part auctions.

SHAI – Smart Hospital AI Operations Platform (Final Year Project)
Python, FastAPI, PostgreSQL, XGBoost, scikit-learn, SHAP, Google Gemini API, Next.js, React, TypeScript, Docker, GitHub Actions

- Built a full-stack AI-powered platform that predicts hospital outpatient no-shows using a calibrated XGBoost model with 40 engineered features and SHAP explanations.
- Built a RESTful FastAPI and PostgreSQL backend with multi-hospital tenancy, consumed by a Next.js dashboard, with Docker and GitHub Actions CI/CD deployment.
- Secured the platform with role-based access (front desk, analyst, manager, admin), scrypt-hashed passwords and signed sessions, with SQLAlchemy and Alembic database migrations and a hospital information system (HIS) sync connector.
- Validated the methodology on 110,000 real appointments (ROC-AUC 0.73 cross-validated, 0.72 on a temporal holdout), with patient-grouped cross-validation and a shuffled-label control (AUC 0.50) confirming no data leakage; deployed model trained on synthetic data pending real hospital data.
- Applied isotonic calibration so risk scores are true probabilities (Brier score about 0.12) and benchmarked XGBoost against logistic regression, decision tree and random forest.
- Built risk triage (High, Medium, Low tiers by clinic capacity) with a priority worklist, plus overbooking optimisation using exact Poisson-binomial calculations, waitlist backfill and ROI simulation.
- Integrated Google Gemini to draft patient reminders in English, Sinhala and Tamil with human review.

SwymWise – Swim Academy Management Mobile App
Flutter, Dart, Riverpod, Supabase, PostgreSQL, TypeScript, Firebase, Stripe

- Built a multi-role mobile platform (swimmer, coach, admin) with training creation, live session tracking, attendance, performance tracking, competitions and a social feed with photo and video posts.
- Designed a serverless backend of about 39 Supabase Edge Functions (Deno/TypeScript) on PostgreSQL with around 90 migrations, using row-level security and multi-tenant data isolation per academy.
- Implemented push notifications with a database-triggered queue, Firebase Cloud Messaging and pg_cron retries, and integrated Stripe subscription billing with idempotent webhook processing.
- Structured the app into 24 feature modules with repository, provider and model layers (about 197k lines of Dart), covered by widget, end-to-end, security and design-system tests.

Other Projects

- **Solo Travel Mate** (Flutter, Dart, Firebase) – Mobile app helping solo travellers in Sri Lanka plan trips with suggested routes, places, pricing, hotels and travel packages.
- **Jumpbook Website Redesign** (Figma) – Redesigned the UI/UX of an online book-selling website and built the prototype in Figma.

EDUCATION

BSc (Hons) in Software Engineering – NSBM Green University, Sri Lanka | 2022 – Present (Final Year)

Relevant modules: Software Engineering, Database Management Systems, Data Structures and Algorithms, Internet and Web Technologies, Information Assurance and Security, Data Science and Analytics, Mobile Application Development.

SOFT SKILLS

Problem Solving, Critical Thinking, Teamwork and Collaboration, Communication, Willingness to Learn, Time Management, Leadership